MERU UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

BACHELOR OF SCIENCE IN DATA SCIENCE

ENHANCING NETWORK SECURITY THROUGH SUPERVISED LEARNING-BASED ANOMALY DETECTION IN NETWORK TRAFFIC

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DECLARATION

This research proposal is my original work prepared with no other than the indicated sources and support, and has not been presented elsewhere for a different or similar assignment.

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DEDICATION

This project is dedicated to everyone who has supported, encouraged and inspired me throughout this journey. To my family, friends and my supervisor I am glad for their unshaken support, and to all who believed in my projects vision your contributions are beyond measure. Thank you for making this possible.

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I would like to express my sincere gratitude to everyone who added any kind of value towards the completion of this project. Special thanks to my mentors for the guidance, my colleagues for their collaboration, and my family and friends for their unwavering support. You all have made this project a reality.

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CHAPTER ONE

INTRODUCTION

1.1 Background of study

Network is a interrelation between entities. A network contains a set of objects and a description of how they are related.(Pang et al., 2021) The simplest network has a minimum of two objects and a relationship that links them. The objects for example may be people and the relationship may be that they are in the same room.

Networks are of many types based on size and scope including the Local Area Networks which covers a small geographical area, Wide Area Networks covers large areas, Metropolitan Area Networks serves as intermediate, and Personal Area Network operate under short distances. All of the above have received a much progressive amount of interest in the past few years. The growth of internet and the availability of affordable computers have made it easier to gather and analyse network data on large quantity (Newman, 2018).

Network topology is the structure in which devices are connected within a network. Some popular topologies include bus, star, ring and mesh.In star all nodes are connected to a central hub, while in ring, they are connected in a circular manner. Whereas for bus, the nodes are connected to a single line of communication, and mesh involves each node being directly connected to several other nodes.

Networks have a set of rules that look over data transmission called protocols, they make sure information is sent and received accurately. The popular and mostly used protocol is the Transmission Control Protocol and Internet Protocol which powers the internet. (Fernandes et al., 2019a) Other protocols include Hypertext Transfer Protocol for web surfing, File Transfer Protocol used for file transfer, and SMTP for email transmission. These protocols ensure compatibility between different systems and smooth communication between networks.

Network use very many devices for communication and traffic control. For instance, routers are used for directing data packets between different networks, making sure they reach their destination. (Partanen et al., 2014) Switches deal with data flow management within one network whereas gateways connect different networks using protocols. Networks also have firewall which acts as security, keeping out unauthorized access to the network, and lastly modems allow devices connect to internet through telephone line.

Due overreliance on networks, security as become a major area. Network security captures protecting data from unauthorized access, attacks from out of the network, and breaches. This have resulted into employing Virtual Private Networks for secure, encrypted connections when on public networks hence securing data from threats.This ensures data integrity and confidentiality which is a primary work of network administrator. Networks also fall into two broad areas, wired and wireless. Wired network is a kind that is connected using physical cables, whereas wireless network allows devices to connect without physical cables, using radio waves for data transmission. Wi-Fi, is one of the most popular types of wireless networking, it enables devices within a certain range to connect to the internet. (Eckenhofer, 2011) Other wireless technologies include Bluetooth, which enables short-range connections, and cellular networks that enable mobile communication over long distances. Wireless networks provide greater flexibility and convenience.

Networking has been pointed out as an important career self-management behaviour, lately research has been limited by traditional forms/channels of communication. (Davis et al., 2020) With growth of social media, networking opportunities have thrived to a new field in technology-based methods. Though not much is known about the positivity of using networking platforms, for example LinkedIn, which was designed for strictly professional profiles. Bringing research together on networking and careers with research on social networking platforms.

Networking is praised by education and marketing experts as a way to adjust business performance a notch higher. Institutions are advised to invest in networking activities, without being able to measure the outcome. (Broad, 2012) Researchers are also following in the markets as networks tradition have identified the know-how of benefits of business networking activities as a field for further studies.

The field of networking is quickly growing, with advancements like 5G, which provides faster internet speeds and lower latency on mobile networks. (Fernandes et al., 2019b)

Another upcoming trend is Internet Of Things, where object are connected to the internet, creating a massive network of smart devices. Software-defined Networking is changing how networks are managed by allowing greater flexibility.

1.2 Motivation for study

Due to the thriving complexity and scale of modern networks, together with the growing need for security and performance optimization. With the rise in cyber threats and demand for high-speed data transmission, understanding and analyzing network traffic has become key for identifying vulnerabilities. This project aims at developing a model for monitoring, analyzing, and optimizing network traffic to improve security and performance.

1.3 The statement of the Problem

Real-life networks tend to be more susceptible to attacks and majorly traffic in them. And networks tend to spread among two interconnected networks which are made up of dependency links and connectivity chains. Traffic in a network is extremely hard to understand hence it leads to traffic load.

To understand the network traffic, it’s important to first know how congestion occurs and have a close look at the network structure which is clearly evident when the network is represented as a graph and the nodes and edges are observed. It’s also wise looking at the ability of the network to handle the attacks and the traffic both topologically and dynamically.

Network administrators find difficulty in avoiding network traffics from occurring within the networks they manage and this is as a result of the fast-growing multimedia applications, such as video streaming, online gaming and high-definition television.

1.4 Research objectives

1.4.1 General objectives

To develop a supervised model that monitors and optimizes network performance to ensure reliability and efficiency in data transfer.

1.4.2 Specific objectives

The objectives of this research project are:

i. To collect network traffic data from public datasets, preprocess it perform feature extraction to find relevant features.
ii. To develop models using multiple algorithms and train them using the preprocessed data.
iii. To evaluate the performance of the developed models and settle on the best performing model.
iv. To create visualizations illustrating the detected anomalies in network traffic.

1.5 Significance of the study

The proposed network traffic analysis system will greatly enhance the security, efficiency, and reliability of modern networks by analyzing traffic patterns. This study addresses gaps in already done research by focusing on real-time anomalies detection, it also enhances analytical models by refining techniques, it helps in real-time threat detection, the study helps in optimizing the performance of a network while reducing costs, finally it increases data privacy enforcement and lead to introduction of a more concrete network security policies.

1.6 Scope of the study

This project will major on capturing, analyzing and interpreting network traffic data to foresee security threats and optimize the networks performance. The study will cover simulated environment, employing various techniques, using a wide range of tools and later maybe used on real-world environment. The study will also look into issues such as data privacy, scalability and limitations of already existing traffic analysis tools. The ultimate objective is to give insights for improving network security and performance.

1.7 Assumptions in the study

In conducting this study, the following assumptions were made. It was assumed that:

1. Network traffic has predictable patterns under normal operating conditions, making it easier to detect anomalies in the traffic.
2. There will be adequate and relevant network traffic data for analysis. Without sufficient data it’s hard to identify trends and patterns.
3. The network traffic data used during the study represents a typical condition within the network under normal conditions.
4. Data collection and traffic monitoring are carried out within the relevant laws and regulations.
5. The traffic data to be used for analysis has not been tampered with so as to mislead analysis.
6. The tools and techniques used for capturing and analyzing network traffic are reliable, accurate, and scalable.
7. The network environment being monitored and analyzed remains stable throughout the study without changes that could affect traffic patterns.

1.8 Limitations of study

Analyzing network traffic involves inspecting packets that my have sensitive data, this raises privacy concerns for instance inspecting corporate network may capture emails.

With the rise in use of encrypted traffic, network traffic analysis maybe limited in its ability to inspect the contents for example a great percentage of web traffic is encrypted making it hard to identify malware.

Real-time traffic analysis requires high computational resources, mostly in large traffic networks for example in a cloud-based environment there is much traffic which could lead to missing some threats.

Traffic analysis systems relying on pattern recognition can give false negative or false positive results, the system can say the traffic is threat free when in real sense there is a threat.

Cyber threats keep developing daily making it hard for the traffic analysis systems to identify the new threats, a zero-day attack might go unnoticed.

Analyzing traffic in a complex network could be exhaustive due to the presence of several layers.

Policies is another limitation regarding data interception and monitoring, in some countries intercepting and analyzing traffic even for security you will need specific approvals.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

In today's world, monitoring and data analysis have become essential for maximizing network efficiency, especially in domains like industrial control systems, optical networks, and the Internet of Things (IoT). Empirical studies demonstrate that the integration of sophisticated machine learning and analytics approaches with real-time data monitoring can significantly improve network visibility, decision-making, and resource management. For example, machine learning-enhanced monitoring systems in optical networks can lead to better resource allocation and a higher quality of service by enhancing traffic engineering, fault detection, and capacity planning (Xu et al., 2019). These developments are critical as networks get bigger and more complicated due to the increased demands for efficiency, security, and dependability.

Multiple studies in the discipline of network security have looked into modern techniques for traffic monitoring and intrusion detection in order to counter new and emerging cyber threats. A reactive monitoring framework, for instance, was created by (Ruff et al., 2021) and can identify anomalies in real-time with little to no effect on system performance. Furthermore, highlighted how deep learning techniques have shown promise in identifying complex traffic patterns, improving monitoring systems' accuracy and efficiency. These developments highlight the ongoing efforts to improve bandwidth efficiency, fortify network security, and lower risks in crucial systems like industrial networks and the Internet of Things. The future will be shaped by the combination of big data, prediction models, and machine learning as research advances.

2.2 Network Traffic Monitoring and Analysis (NTMA)

(Ruff et al., 2021) concentrate on using deep learning algorithms for network activity monitoring and analysis. In their 2019 study, (D'Alconzo et al., 2019). explore the function of large data sets in NTMA, emphasizing in particular techniques for handling and evaluating massive amounts of network traffic. An overview of the several methods applied to network data analysis and forecasting is given by. The application of analytics and monitoring in optical networks is discussed by (Hwang et al., 2020), who place a focus on real-time monitoring and associated designs. Collectively, these publications present the developing approaches to network traffic management and analysis using contemporary technologies.

2.3 Intrusion Detection and Network Security

(Chaabouni et al., 2019) highlight the intricacy of IoT networks as they investigate learning strategies for network detection of attacks in IoT contexts. For industrial networks, (Liang et al., 2019) provide a network surveillance technique that is optimized using multifeatured data clustering. A taxonomy of network risks is presented, and also examine how datasets affect intrusion detection systems. (Liu et al., 2020) investigate methods for handling unbalanced traffic on networks in systems for intrusion detection using machine learning along with deep learning. In order to increase network security deep network structures for identify breaches while concentrating on original flow data.

2.4 Network Optimization

(Hwang et al., 2020) offer insightful analyses of a range of network optimization-related topics.He also emphasize effective reactive monitoring strategies to boost utilization of resources and reaction times in order to improve network performance. (Ali et al., 2020) talk about elastic and dependable bandwidth reserve schemes that maximize bandwidth allocation through distributed load monitoring and control techniques. In the setting of the Internet of Things (IoT), (Ali et al., 2020) investigate network optimization, with a focus on resource management and energy efficiency enhancement. When taken as a whole, these studies help to clarify how to improve networks in a variety of contexts, from general the network's performance to difficulties unique to the Internet of Things.

2.5 Traffic Generation and Data Flow

(Vikram, 2020) concentrate on the structural study of network data flows, offering understanding into the characterization and analysis of traffic patterns. Their work emphasizes how crucial it is for network managers to comprehend flow structures. In their investigation of generative adversarial networks (GANs) for traffic creation, (Vikram, 2020) show how well flow-based models can mimic network traffic. Their methodology demonstrates how traffic generation tactics can be improved by machine learning approaches. By providing innovative techniques for capturing and simulating network traffic patterns, both works advance the discipline of traffic analysis.

2.6 Mobile Ad Hoc and Industrial Networks

(Carrera et al., 2022) provide a dependable data transmission scheme for mobile ad hoc networks (MANETs) that uses signcryption. Their method combines signature and encryption techniques to improve data safety and efficiency in MANETs. (Carrera et al., 2022) concentrates on protecting networks of vital infrastructure, especially for industrial control systems and Supervisory Control and Data Acquisition (SCADA) systems. He talks about the flaws in these important systems and stresses the importance of strong security. The security and dependability of interconnected systems in mobile and industrial contexts are greatly improved by both works.

2.7 Research Gaps

2.7.1 Lack of scalability in machine learning and deep learning models

The increase in amount and velocity of data might cause scalability problems when applying deep and machine learning models for network traffic monitoring. This can impede real-time analysis and overwhelm conventional systems. The capacity to detect and respond to threats in real time can be greatly improved by resolving scalability concerns in deep learning models for network traffic monitoring, which will improve cybersecurity overall.

2.7.2 Big data analysis in real-time

An developing subject that focuses on analyzing and processing the massive amounts of data created by network operations in real time is real-time massive data analytics for network traffic monitoring. By seeing trends, abnormalities, and possible threats early on, this field of study seeks to improve network performance, security, and dependability. By promptly identifying and mitigating any harm, companies can improve the security of their networks by finding solutions to real-time big data analytic problems related to network traffic monitoring.

2.7.3 Lack of better intrusion detection algorithm

The state of detection of breaches algorithms today finds it difficult to effectively handle the special difficulties presented by heterogeneous and resource-constrained networks. The limited computational capacity and diverse device capabilities present in these environments pose challenges to the efficient operation of conventional algorithms. Developing efficient intrusion detection techniques for heterogeneous and resource-constrained networks greatly improves overall network security.

2.7.4 Lack of new methods to analyze traffic flows in highly distributed networks

The dearth of novel approaches makes traffic flow analysis in massively distributed networks extremely difficult. The intricacy and dynamism of contemporary networks are frequently too much for traditional methods, which results in inefficient data handling and analysis. Therefore, the creation of sophisticated analytical techniques that are capable of accurately capturing and deciphering the complicated patterns of movement of traffic in such contexts is desperately needed.

2.8 Summary

Monitoring, data analysis, and complex network environments are essential for enhancing work efficiency, safety, and resource management. Particularly in control systems for industries and IoT networks, analytics and machine learning improve real-time monitoring by helping with activities like problem identification, capacity planning, and traffic engineering. Improved intrusion detection systems improve network security, however there are still issues with scalability, actual time massive data analytics, and algorithmic limits. In distributed networks, traffic flow analysis demands novel methods to deal with dynamic data and growing complexity. In order to improve network security, performance, and optimization, these holes must be filled.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

An already existing dataset from Kaggle will be used to examine network activity in this network traffic research project. The collection includes comprehensive network traffic logs that include timestamps, packet information, and protocol specifics. Networking protocols will be decoded using protocol analysis in order to analyze data flow and find any possible irregularities or security concerns. The dataset's patterns, trends, and anomalies will be found using statistical techniques including time-series analysis, which provided information on bandwidth consumption, unusual traffic, and other important network performance metrics. Choosing a Kaggle dataset enables a dependable and varied compilation of actual network traffic data, which is essential for evaluating theories and spotting possible weaknesses. While statistical analysis is crucial for spotting trends and odd traffic patterns to enhance network performance or identify possible security threats, protocol analysis is still necessary for analyzing protocol-specific behaviors and identifying any violations or inefficiencies in data flow. When combined, these techniques offer a strong foundation for comprehending network traffic without requiring real-time packet capture, which makes the analysis feasible and expandable.

3.2 Research Design

3.2.1 Type Of Study

With a few exploratory components, this network traffic analysis project falls under the category of descriptive research. Because it uses the given Kaggle dataset to evaluate and characterize current trends in network traffic, it is descriptive. Finding patterns, behaviours, and abnormalities in the data is the main goal in order to enhance comprehension of network security and performance (Wong & Arjunan, 2024). The study will use the exploratory component to look for potentially new patterns or risks in the dataset that haven't been found before.

3.2.2 Research Framework

The purpose of this study's research approach is to characterize the features of network traffic, including anomalies, protocol distribution, and bandwidth utilization. With an emphasis on characteristics like IP addresses, port numbers, timestamps, protocols, and packet sizes, information will be gathered from the Kaggle network traffic dataset. Statistical analysis will be performed and visualization of traffic trends over time along with protocol usage utilizing heat maps, box graphs, and histograms. To find various traffic kinds or possible anomalies, clustering or classification algorithms will be used. In order to evaluate general patterns and spot anomalous behaviours like DDoS attacks, the findings will be displayed using summary statistics and infographics. In order to provide a solid basis for a descriptive study of network traffic data, the conclusions will address observed network patterns and its effects for security or performance optimization.

3.3 Data Collection Methods

3.3.1 Description Of Data Sources

The dataset to be used in this research came from Kaggle, a website that houses a large collection of publicly accessible datasets that are mostly categorized as secondary data. Researchers can examine pre-existing datasets without having to gather new data by using secondary data, which is information that has already been gathered and made public by others. Numerous network traffic-related variables, including IP addresses, timestamps, packet sizes, and protocols, are included in the Kaggle dataset.

3.3.2 Techniques Used

Techniques like data downloading and data cleansing will be used for data retrieval. The dataset, which is available in CSV and JSON formats on the Kaggle platform (Wong & Arjunan, 2024). The dataset will be prepared for analysis by doing preliminary data cleaning after it was downloaded to address any missing values, discrepancies, or outliers. Instead of using techniques like surveys, online scraping, or API data retrieval to obtain new information, this strategy shortened the process and allowed for an emphasis on gleaning insights from the available data (Wong & Arjunan, 2024).

3.4 Data Preparation

3.4.1 Data Cleaning Methods And Handling Missing Values

To guarantee the accuracy and dependability of the data, a number of data cleaning techniques will be used. Since missing entries could distort the analysis, managing missing data is a crucial first step. In order to remedy this, the impact of missing values on the dataset will be evaluated. Rows will be eliminated if the percentage of missing data for a given feature will be negligible. Imputation strategies, such as employing the mode for categorical variables or the median or mean for numerical attributes, are to be used for features with a large quantity of missing data.

3.4.2 Data Transformation Procedures

Procedures for data transformation will be employed to improve the dataset's analytical effectiveness. This involved normalizing features to guarantee that they contribute equally during analysis, converting categorical variables into numerical values using methods like one-hot encoding, and standardizing formats for timestamp data to guarantee consistency across the dataset. In order to prevent skewing the results, outliers will also be discovered using statistical techniques. When combined, these data transformation and cleaning methods to produce a more polished dataset that made network traffic analysis more accurate and insightful (Hooshmand & Hosahalli, 2022).

3.5 Data Analysis Techniques

3.5.1 Statistical Methods

The dataset will be summarized using statistical analysis methods including descriptive statistics, which include measures of spread (standard deviation, variance, and interquartile range) and central tendencies (mean, median, and mode). The data distribution will be examined for patterns and abnormalities using visualization techniques such as scatter plots, box plots, and histograms.

3.5.2 Machine Learning Algorithms

Various algorithms will be used for machine learning, depending on the objectives of the investigation. Algorithms like Support Vector Machines (SVM), Random Forests, and

Logistic Regression will be used for classification tasks in order to identify abnormalities and categorize various kinds of network traffic. In order to group comparable traffic patterns and make it easier to identify anomalous activity that could be a sign of security risks, KMeans was used for clustering (Wang et al., 2021).

3.5.3 Tools And Software

Python will be used for the entire investigation, utilizing a variety of modules and tools that facilitate statistical analysis, machine learning, and data manipulation. Matplotlib and Seaborn will offer strong visualization capabilities, while libraries like Pandas and NumPy are to be utilized for data cleaning and manipulation. Scikit-learn, which provides a large number of algorithms and tools for model training, evaluation, and performance metrics, will be the main package used for machine learning applications. All things considered, the network traffic information will be thoroughly analysed due to the combination of statistical techniques, machine learning algorithms, and Python libraries, which is to make it possible to successfully identify patterns and abnormalities.

3.6 Model Building

3.6.1 Model Selection

K-Nearest Neighbors (KNN), Random Forest, and Support Vector Machine (SVM) will all be taken into account for selection. Each method will be used on the dataset, and its results will be evaluated based on the accuracy of classification in order to determine which model is the most effective.

3.6.2 Training And Testing Phase

To assess the generalization performance of the Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) models, a subset of the dataset (usually 70–80%) will be used for training, and the remaining data will be used for testing. In order to minimize variance and prevent overfitting, the Random Forest model will build an ensemble of decision trees during training, each trained on a random subset of characteristics and data samples. In order to maximize the margin between data points from various classes, the SVM model will employ a hyperplane to divide classes in the feature space (Wang et al., 2021). In the meanwhile, KNN will use the most popular label among the K nearest neighbors to categorize data points.

Following training, unseen testing data will be used to assess each of the three models.

3.6.3 Evaluation Metrics

Precision, recall, F1 score, and accuracy are the evaluation measures that will be used to evaluate the Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) models' performance. Each model will be assessed according to its accuracy (the capacity to predict positive instances), recall (the ability to accurately determine true positive cases), and F1 score, which is the balance of these two metrics. Furthermore, overall precision will be taken into account in order to assess the models' general prediction ability. The choice of the best model for the job was aided by the comparison of these measures, which provided information about the advantages and disadvantages of each approach.

3.7 Ethical Considerations

3.7.1 Data Privacy and Security

Network traffic analysis involves sensitive data, data privacy and security are important ethical considerations. Network traffic includes information on users' communications and online activity, as well as potentially private data like IP addresses, browsing preferences, and even message content. Network traffic analysis may unintentionally reveal private or personally identifiable data (PII) if privacy is not carefully considered. This could result in abuse or violations of privacy rights.

The core of this project is to have an ethical obligation to protect user privacy by putting policies in place including data anonymization, encryption, and access control for sensitive data. While encryption shields data in transit from illegal access or interception, anonymization makes sure that specific users cannot be readily identified from the data being examined. To guard against potential security flaws, it's also essential to establish stringent guidelines on data usage and make sure that only authorized individuals have access to network data.

Addressing privacy issues also requires transparency. When their data is being gathered, why it is being analyzed, and how it will be utilized, kept, and safeguarded should all be disclosed to users. The analysis is further guaranteed to be carried out responsibly and legally by obtaining consent where required and according to pertinent legislative frameworks, such as the General Data Protection Regulation (GDPR).

Ethical considerations in security include safeguarding the systems from which the data is gathered as well as the data being examined. This project will reveal system flaws or vulnerabilities that, if left unchecked, could be exploited. Any findings pertaining to security flaws must be handled appropriately in a manner that reduces risks without making them worse (Carrera et al., 2022).

3.7.2 Ethical Implications Of This Research

Privacy Concerns

There are privacy concerns because this project will collect sensitive or personal data, such as location data, login activity, and browsing history. Individual privacy rights may be violated by improper management of this data, hence it is crucial to handle these concerns carefully. When possible, getting user agreement should be part of ethical traffic analysis procedures; nevertheless, in situations like corporate networks, where this could be less practicable, transparency standards are essential. These guidelines encourage trust and responsibility in data handling by educating users about monitoring procedures.

Potential For Misuse

This project has the potential to be misused since it might be used to track and regulate user activity beyond what is required by security measures, which could result in profiling based on network behaviors or the needless tracking of individual actions. Malicious actors can also take advantage of this data if it is processed or stored in an unsafe manner, which leaves it open to illegal access. Users may suffer fraud, identity theft, and other consequences as a result of such breaches. Therefore, to avoid abuse and safeguard user privacy, cautious handling and secure protocols are crucial.

Transparency And Accountability

The techniques used in this project to identify patterns may be opaque, therefore it is crucial to make sure they are clear and intelligible, especially when they have an impact on choices regarding user access or security. If the algorithms or techniques exhibit biases, such as unjustly profiling particular user groups, ethical issues can surface. Fairness and inclusion are therefore necessary to develop equitable tools. For network traffic analysis to be done responsibly, algorithm design must be transparent and accountable.

Data Minimization

Ethical data practices prioritize data reduction, which is the process of gathering only the information required to achieve a particular goal. Restricting data gathering lowers security risks and stops data from being misused for unforeseen uses. Excessive data collection can reveal private information and raise the possibility of moral dilemmas. Organizations protect privacy and improve data security by concentrating on data reduction.

3.8 Limitations

3.8.1 Potential Biases

Dataset Bias: Publicly accessible datasets, which might not accurately reflect all traffic patterns or behaviors (such as attack types or geographic locations), are frequently used in network traffic analysis. For instance, statistics from particular organizations or nations may show distinct traffic patterns that aren't applicable to other contexts. This may result in algorithms that function well in one situation but poorly in another.

Selection Bias: Models may become biased towards particular forms of traffic if they are over- or under-represented in the training dataset (e.g., malicious vs. normal traffic). Because of this, certain types of traffic may be either overreported or underreported, resulting in skewed detection rates.

Observer Bias: Based on institutional guidelines or personal knowledge, researchers may inadvertently overlook some aspects of the traffic while focusing on others they believe to be important. This may make it more difficult to identify novel or uncommon traffic kinds that don't fit established patterns.

Infrastructure Bias: Because data flows differently over wired and wireless networks, for example, studies that examine traffic utilizing particular network infrastructure types may produce disparate findings. Because of this, extrapolating results to different network architectures may be challenging.

Confirmation Bias: Its the tendency for researchers to choose models or results that validate their original theories. This can cause network traffic analysis to ignore other explanations or methods in favor of choosing models or adjusting parameters that support preconceived notions about traffic patterns.

Labeling Bias: Manually classifying network traffic as benign or malicious can be arbitrary or inconsistent, especially when it comes to differentiating between gray areas. Inaccuracies in supervised models may result from variations in labeling procedures or from a lack of uniformity.

3.8.2 Limitations In Data

Restricted Dataset Scope: Because they only include information from a small number of networks, settings, or traffic kinds, many network traffic datasets (like those on Kaggle) may not be very diverse. It is challenging to apply results globally due to this lack of generalization.

Incompleteness: The capacity to test a broad range of scenarios may be hampered by public datasets' lack of data from particular scenarios (for example, insufficient data on particular attack types or low prevalence of uncommon traffic patterns).

Aging Data: The applicability of results to current networks may be limited by the presence of out-of-date protocols or network behaviors in public datasets that do not fairly depict contemporary traffic.

Data Volume Constraints: Due to storage and processing resource restrictions, network traffic data may be too large, which may cause sampling or downsampling and possibly miss important but uncommon events.

Inadequate Labeling: It can be difficult to precisely identify harmful actions when labeling network traffic data. Inaccuracies and noise introduced by poor labeling may have an impact on model training and performance.

3.8.3 Methodological Limitations

Feature Selection: Because of the enormous dimensionality and complexity of data, it might be difficult to choose relevant characteristics from unprocessed network traffic. Biased or insufficient analysis may result from poor or inaccurate feature selection.

Overfitting: Deep learning and other machine learning models have the potential to overfit data, identifying noise instead of actual traffic patterns. Models that perform well on training data but poorly in real-world situations may result from this overfitting.

Model Transferability: Due to the vast differences in traffic patterns, models that were trained on a single dataset or network environment might not function effectively in other contexts. Scalability of solutions is limited by a lack of transferability.

Managing Unbalanced statistics: There are frequently imbalances in network traffic statistics, with benign traffic far outnumbering malicious traffic. It is challenging to address this imbalance, which can distort machine learning models' performance.

Detecting Secretive Techniques Is Difficult: Attackers employ more complex methods to avoid detection, such as encrypted or fragmented communications. Because of their limited adaptability, standard techniques may not be able to detect such deceitful activities.

Limitations on Real-time Processing: Since many models are developed for offline analysis, they might not be able to process network traffic in real-time, which is essential for threat identification and response.

3.9 Summary

Recap Of Key Methodological Steps

Comprehensive logs from an existing Kaggle dataset are used in this project, enabling protocol decoding and anomaly detection to investigate bandwidth usage and questionable activity patterns. To identify trends, classify traffic, and identify potential security concerns, the dataset will be analyzed using statistical techniques, time-series analysis, and visualizations such as heat maps. Traffic classification and anomaly detection will be aided by machine learning techniques like SVM, Random Forest, and KNN; model selection is guided by performance metrics like accuracy, recall, and F1 score. This method preserves data privacy and security compliance by enabling scalable analysis without real-time packet capture through the use of powerful data cleaning, transformation, and Python tools.

Justification Of Chosen Methods

The necessity for realistic and varied data, which the Kaggle dataset provides and enables the examination of real-time network traffic patterns without the need for packet capture, justifies the research methodologies that were used. While statistical techniques reveal important trends like bandwidth utilization and odd traffic patterns that can point to vulnerabilities, protocol analysis offers crucial insights into network behaviors that make it possible to identify security or performance flaws particular to a given protocol. Descriptive statistics and machine learning algorithms efficiently categorize traffic and identify irregularities, providing a comprehensive perspective of network activity and security. When combined, these techniques provide a strong and adaptable framework for network traffic analysis, which is essential for identifying possible risks and improving network performance in general.

CHAPTER FOUR
RESULTS AND DISCUSSION

4.1 Introduction

In this chapter, the process of detecting network anomalies using machine learning techniques is explored. The chapter begins with an overview of the dataset used, followed by a detailed explanation of the data pre-processing steps. The chapter then discusses the development and evaluation of various machine learning models, including Logistic Regression andju Support Vector Machine (SVM). The results of the model evaluations are presented, and the performance of each model is analysed. Finally, the chapter concludes with a discussion of the findings and their implications for network security.

4.2 Dataset Overview

42 features total, including protocol type (e.g., tcp, udp, icmp), service (e.g., http, ftp, telnet), connection status (e.g., SF, S0, REJ), and other statistical traffic features (e.g., duration, bytes transmitted, error rates) are included in this collection of network traffic records. A target variable specifying the type of traffic, such as "normal" or specific assaults (e.g., neptune, ipsweep, portsweep, smurf, and warezclient), is tagged on each of the 1,260 entries in the dataset. The dataset is intended for use in machine learning-based tasks including intrusion detection and network anomaly detection.

4.3 Loading Libraries and Dataset

Two essential Python libraries for numerical computations and data processing are imported: NumPy and Pandas. The os.walk() method is then used to navigate across the directory "C:/Users/Ouma Becks/Desktop/NAD" and list all of the filenames that are present.

```python
import numpy as np
import pandas as pd

import os
for dirname, _, filenames in os.walk('"C:/Users/Ouma Becks/Desktop/NAD"'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

Figure 4-1

The required libraries for data processing, visualization, machine learning, and assessment are imported. NumPy and Pandas handle numerical computations and data management, while Seaborn and Matplotlib assist with data display. Multiprocessing and itertools provide tools for performance enhancement and iteration, respectively, while warnings stop unnecessary alarms. The script divides the dataset using train_test_split from Scikit-Learn, selects features using RFE, then preprocesses using StandardScaler and LabelEncoder. Machine learning models such as Support Vector Machine (SVM) and Logistic Regression are imported for categorization challenges. Finally, several assessment metrics, such as accuracy_score, confusion_matrix, and roc_curve, are included to gauge the model's performance.

```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import time
import multiprocessing
import itertools
import warnings

from collections import Counter
from pandas.api.types import is_numeric_dtype

# Machine Learning Models
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.feature_selection import RFE
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC, LinearSVC as svm

# Metrics & Evaluation
from sklearn.metrics import (
    accuracy_score, confusion_matrix, precision_score, recall_score,
    f1_score, roc_auc_score, classification_report, precision_recall_curve,
    r2_score, mean_squared_error, roc_curve
)

# Suppress warnings
warnings.filterwarnings("ignore")
```

Figure 4-2

4.3.1 Loading the dataset

The first five rows of the data DataFrame below show the structure of the dataset, including its columns and sample values.

	0	tcp	private	REJ	0.1	0.2	0.3	0.4	0.5	0.6	...	0.04.1	0.06.1	0.00.3	0.00.4	0.00.5	0.00.6	1.00.2	1.00.3	neptune	21
0	0	tcp	private	REJ	0	0	0	0	0	0	...	0.00	0.06	0.00	0.00	0.00	0.0	1.00	1.00	neptune	21
1	2	tcp	ftp_data	SF	12983	0	0	0	0	0	...	0.61	0.04	0.61	0.02	0.00	0.0	0.00	0.00	normal	21
2	0	icmp	eco_i	SF	20	0	0	0	0	0	...	1.00	0.00	1.00	0.28	0.00	0.0	0.00	0.00	saint	15
3	1	tcp	telnet	RSTO	0	15	0	0	0	0	...	0.31	0.17	0.03	0.02	0.00	0.0	0.83	0.71	mscan	11
4	0	tcp	http	SF	267	14515	0	0	0	0	...	1.00	0.00	0.01	0.03	0.01	0.0	0.00	0.00	normal	21

5 rows × 43 columns

Figure 4-3

4.3.2 Data information

The data.info() function provides a quick summary of the dataset, together with information on column names, data types, memory usage, and the number of non-null objects..

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 22543 entries, 0 to 22542
Data columns (total 43 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   0        22543 non-null  int64
 1   tcp      22543 non-null  object
 2   private  22543 non-null  object
 3   REJ      22543 non-null  object
 4   0.1      22543 non-null  int64
 5   0.2      22543 non-null  int64
 6   0.3      22543 non-null  int64
 7   0.4      22543 non-null  int64
 8   0.5      22543 non-null  int64
 9   0.6      22543 non-null  int64
 10  0.7      22543 non-null  int64
```

Figure 4-4

4.3.3 Data shape

There are 43 columns and 22,543 rows in the dataset. A single observation or data point is represented by each row, while a distinct feature is represented by each of the 43 columns.

```
(22543, 43)
```

4.3.4 Data description

Below is a statistical summary of all the numerical columns in the dataset. It includes the count, which shows the number of non-null values in each column, and the mean, which is the average value. The standard deviation (std), which displays the distribution of data, aids in understanding variability. The minimum (min) and maximum (max) values indicate the range of data for each attribute.

count	22543.000000	2.254300e+04	2.254300e+04	22543.000000	22543.000000	22543.000000	22543.000000	22543.000000	22543.000000	22543.000000
mean	218.868784	1.039591e+04	2.056110e+03	0.000311	0.008428	0.000710	0.105399	0.021648	0.442222	0.119904
std	1407.207069	4.727969e+05	2.121976e+04	0.017619	0.142602	0.036474	0.928448	0.150331	0.496661	7.269758
min	0.000000	0.000000e+00	0.000000e+00	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000e+00	0.000000e+00	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	5.400000e+01	4.600000e+01	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	2.870000e+02	6.010000e+02	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000	0.000000
max	57715.000000	6.282565e+07	1.345927e+06	1.000000	3.000000	3.000000	101.000000	4.000000	1.000000	796.000000

8 rows × 39 columns

Figure 4-5

4.4 Data Preprocessing

To guarantee the best possible model performance, a number of pre-processing methods were used:

4.4.1 Handling missing values

There was no need for imputation because the dataset was examined for missing values and none were discovered.

```
0          0
tcp        0
private    0
REJ        0
0.1        0
0.2        0
0.3        0
0.4        0
0.5        0
0.6        0
0.7        0
0.8        0
0.9        0
0.10       0
```

Figure 4-6

4.4.2 Encoding categorical features

Label Encoding was used to translate the category variables (flag and service) into numerical representations. For machine learning models that don't directly handle categorical input, this modification was necessary..

	0	tcp	private	REJ	0.1	0.2	0.3	0.4	0.5	0.6	...	0.04.1	0.06.1	0.00.3	0.00.4	0.00.5	0.00.6	1.00.2	1.00.3	neptune	21
0	0	1	45	1	0	0	0	0	0	0	...	0.00	0.06	0.00	0.00	0.00	0.0	1.00	1.00	14	21
1	2	1	19	9	12983	0	0	0	0	0	...	0.61	0.04	0.61	0.02	0.00	0.0	0.00	0.00	16	21
2	0	0	13	9	20	0	0	0	0	0	...	1.00	0.00	1.00	0.28	0.00	0.0	0.00	0.00	24	15
3	1	1	55	2	0	15	0	0	0	0	...	0.31	0.17	0.03	0.02	0.00	0.0	0.83	0.71	11	11
4	0	1	22	9	267	14515	0	0	0	0	...	1.00	0.00	0.01	0.03	0.01	0.0	0.00	0.00	16	21

Figure 4-7

4.4.3 Duplicates

The dataset's total number of duplicate rows is displayed below. Every record is assured to be unique since 0 denotes the absence of duplicate entries.

```
print(f"Number of duplicates: {data.duplicated().sum()}")
```

```
Number of duplicates: 0
```

Figure 4-8

4.4.4 Selecting categorical features

Printed is the summary categorical features of the data.

	protocoltype	service	flag	attack
0	tcp	ftp_data	SF	normal
1	udp	other	SF	normal
2	tcp	private	S0	neptune
3	tcp	http	SF	normal
4	tcp	http	SF	normal

Figure 4-9

4.4.5 Generalizing target feature

Changes the DataFrame df's 'attack' column by substituting the string 'attack' for any value that is not 'normal'.

```
df['attack'].loc[df['attack']!='normal']='attack'
```

Figure 4-10

4.4.6 Encoding the above categorical features

Protocoltype, service, flag, and attack are categorical columns that can be converted into numerical values using LabelEncoder() so that machine learning algorithms can use them..

```
le=LabelEncoder()
df['protocoltype']=le.fit_transform(df['protocoltype'])
df['service']=le.fit_transform(df['service'])
df['flag']=le.fit_transform(df['flag'])
df['attack']=le.fit_transform(df['attack'])
```

Figure 4-11

4.4.7 Correlation matrix

The heatmap has a headline ("Correlation Matrix of NAD") for context, shows correlation values adjusted to one decimal point, and employs a blue-green color gradient (YlGnBu).

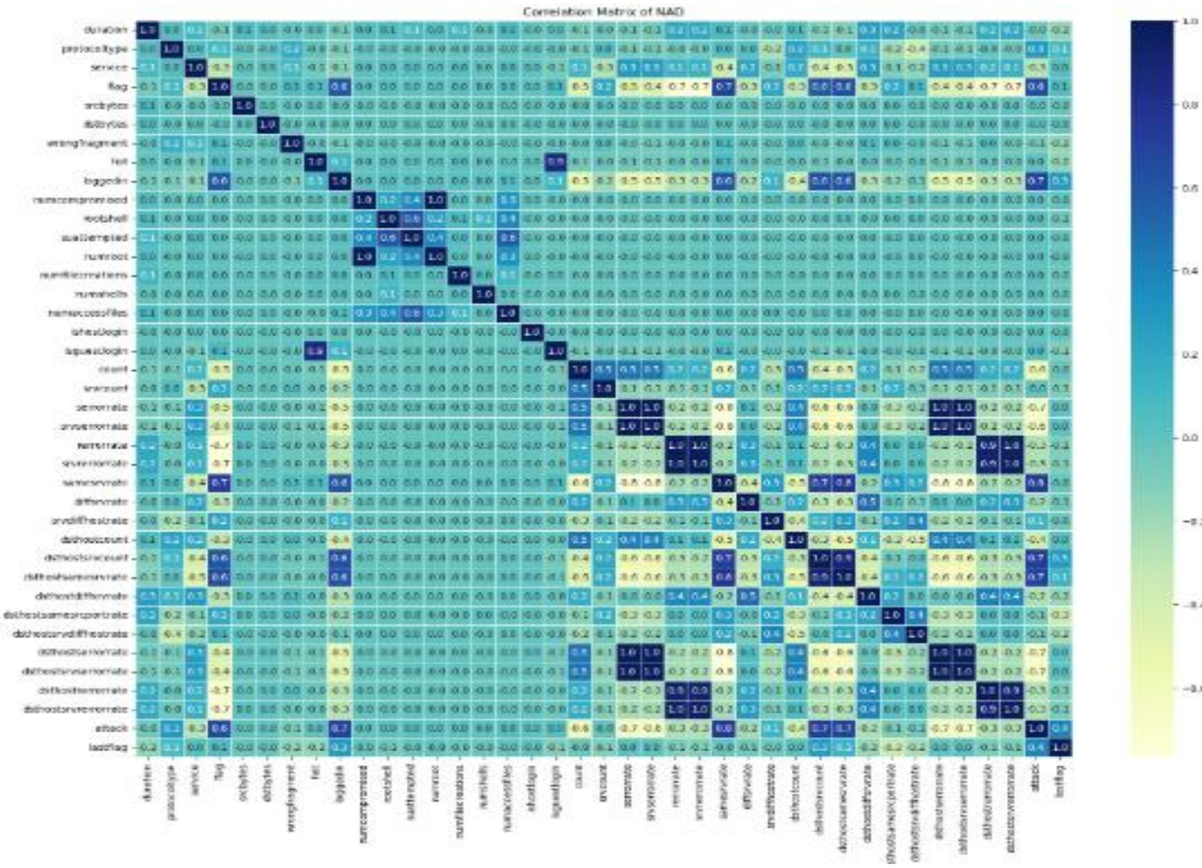


Figure 4-12

4.5 Features Selection

4.5.1 Filtering features according to the heatmap

To see their pairwise correlations, a heatmap is created after first filtering a correlation matrix to only contain the designated characteristics ('duration','srcbytes', 'dstbytes', 'count','srvcount', and 'attack') from the DataFrame df.

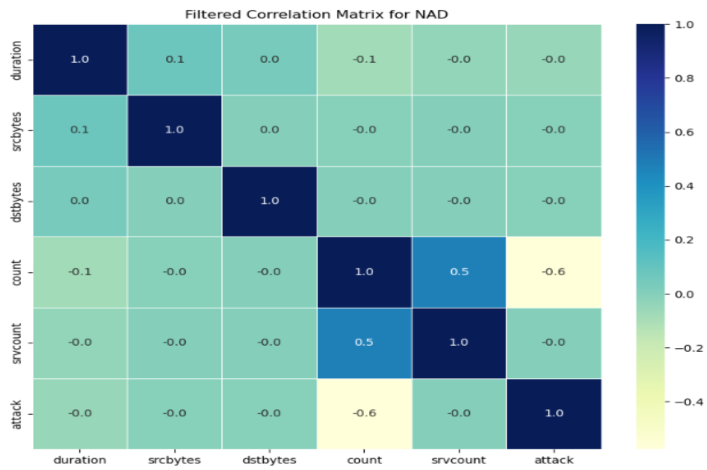


Figure 4-13

Separating X and Y features

Removes the "attack" column from X while leaving it as y, separating the feature variables (X) and the target variable (y) from the DataFrame df.

```
X=df.drop(['attack'],axis=1)
y=df['attack']
```

Figure 4-14

4.5.2 Displaying the target distribution

uses the Counter function from the collections module to count the instances of each distinct class label in y and outputs the class distribution of y.

```
Class distribution: Counter({1: 67343, 0: 58630})
```

Figure 4-15

4.5.3 Splitting the dataset

Divides the dataset into subsets for testing and training, allocating 30% of the data for testing (X_test, y_test) and 70% for training (X_train, y_train). Reproducibility is ensured by random_state=42, which means that the split stays constant throughout several runs.

```
# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

Figure 4-16

4.5.4 Selecting training features

Using the ANOVA F-test, which assesses the statistical association between each feature and the target variable, the top ten most pertinent features are chosen from the dataset. The names of the selected features that are anticipated to be the best predictive in the specified classification job are then printed after applying this selection to the training and test datasets.

```
Selected Features: Index(['flag', 'loggedin', 'count', 'serrorrate', 'srvserrorrate',
       'samesrvrate', 'dsthostsrvcount', 'dsthostsamesrvrate',
       'dsthostserrorrate', 'dsthostsrvserrorrate'],
      dtype='object')
```

Figure 4-17

4.6 Model Training and Selection

These two Supervised machine learning models, Support Vector Classification (SVC), and Logistic Regression, were trained and evaluated.

4.6.1 Logistic regression

With an accuracy of 91.56%, the logistic regression model performed well overall. With both classes having excellent precision (93% for attack, 90% for normal) and recall (89% for attack, 94% for normal), the classification report demonstrates that the model does a good job of differentiating between "attack" and "normal" traffic, producing balanced F1-scores of roughly 91–92%.

```
============================== Logistic Regression ==============================

Logistic Regression Accuracy: 0.9156435224386114


Logistic Regression Classification Report of NAD:

              precision    recall  f1-score   support

      attack       0.93      0.89      0.91     17709
      normal       0.90      0.94      0.92     20083

    accuracy                           0.92     37792
   macro avg       0.92      0.91      0.92     37792
weighted avg       0.92      0.92      0.92     37792
```

Figure 4-18

4.6.2 Support Vector Machine (SVC)

Used the SVC() class to train a Support Vector Machine (SVM) model. The model is fitted using the matching labels (y_train) and the converted training dataset (X_train_transformed). The model's accuracy is determined by comparing its predictions with the actual test labels (y_test) after training on the converted test dataset (X_test_transformed). A classification report, which offers metrics like precision, recall, and F1-score for several classes, is used to further assess the classification performance.

```
============================== Support Vector Machine (SVM) ==============================

SVM Accuracy: 0.9031805673158341


SVM Classification Report of NAD:

              precision    recall  f1-score   support

      attack       0.89      0.91      0.90     17589
      normal       0.92      0.90      0.91     20203

    accuracy                           0.90     37792
   macro avg       0.90      0.90      0.90     37792
weighted avg       0.90      0.90      0.90     37792
```

Figure 4-19

4.7 Model Evaluation

The performance of each model was assessed using key metrics:

4.7.1 Accuracy

Logistic Regression

With an accuracy of 91.56%, the logistic regression model performed well.

```
Logistic Regression Accuracy: 0.9156435224386114
```

Figure 4-20

Support Vector Machine

With an accuracy of 90.31%, the SVM model demonstrated excellent performance in differentiating between attack and legitimate traffic.

```
SVM Accuracy: 0.9031805673158341
```

Figure 4-21

4.7.2 Precision, Recall, Support and F1-score

Logistic Regression

Both classes have balanced F1-scores of roughly 91–92% due to their strong precision (93% for attack, 90% for normal) and recall (89% for attack, 94% for normal).

	precision	recall	f1-score	support
attack	0.93	0.89	0.91	17709
normal	0.90	0.94	0.92	20083

Figure 4-22

Support Vector Machine

The attack and normal f1-scores (0.90 and 0.91, respectively) show a strong balance between recall and precision.

	precision	recall	f1-score	support
attack	0.89	0.91	0.90	17589
normal	0.92	0.90	0.91	20203

Figure 4-23

4.7.3 Confusion matrix analysis

Logistic Regression

To display the distribution of accurate and inaccurate predictions, a confusion matrix is calculated and displayed using a heatmap. The assault (0) and normal (1) classes are distinguished by the plot labels. Following the extraction of the confusion matrix values, the number of true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP) is displayed.

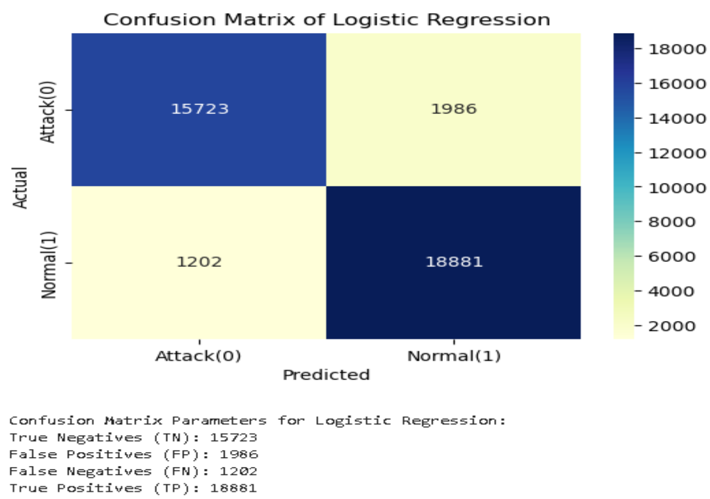


Figure 4-24

Support Vector Machine

According to the confusion matrix, 1,600 normal samples were incorrectly identified as assaults (FP), but 15,989 attack samples were correctly classified (TN). Similarly, 18,144 normal samples were correctly identified (TP) while 2059 attack samples were incorrectly labeled as normal (FN).

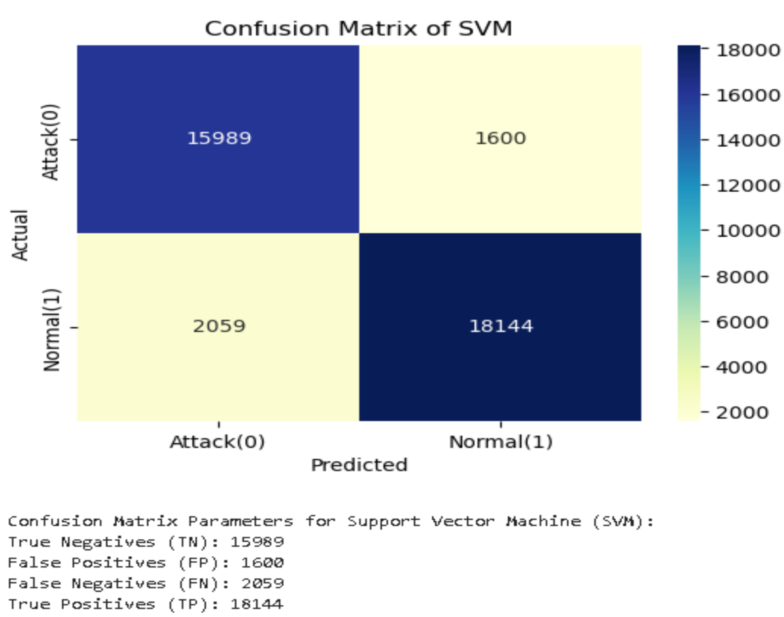


Figure 4-25

The findings show that Support Vector Classification (SVC) has the best performed out of the two models that were put into use.

4.8 Visualization

4.8.1 ROC curves

The Support Vector Machine (SVM) and Logistic Regression (LR) models' classification performance is compared in the shown ROC curves, and each model's AUC value shows how well it can differentiate between classes. If one model's curve is continuously above the other (Logistic Regression), it shows superior anomaly detection capability; conversely, curves near the diagonal (SVM) indicate weak predictive power. A higher AUC indicates better model performance.

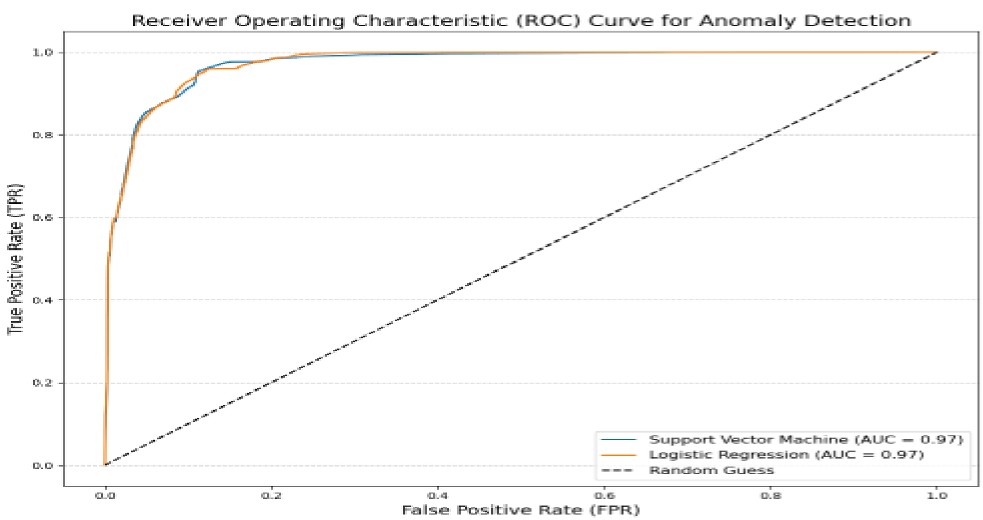


Figure 4-26

4.8.2 Anomaly score distribution

With the SVM plot on the left and the Logistic Regression plot on the right, the two KDE plots display the anomaly score distributions for both normal and attack events. The threshold is shown by a vertical dashed line in each figure, which makes it easier to see how normal and attack occurrences differ according to their individual anomaly scores.

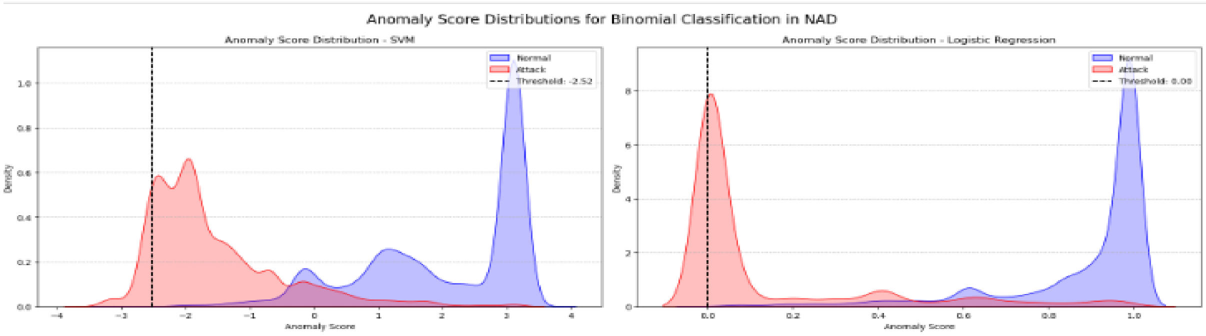
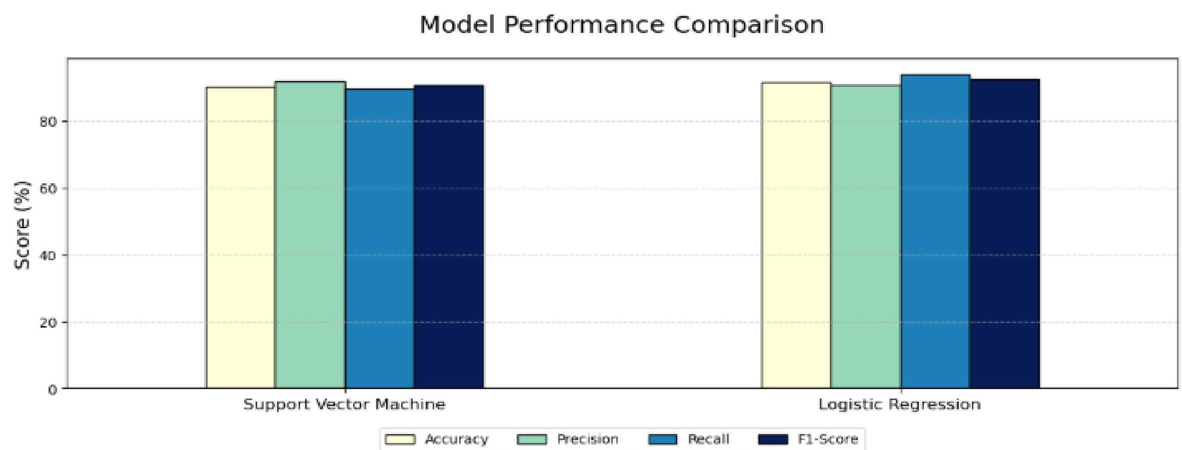


Figure 4-27

4.8.3 Model performance summary

A visual representation of the model evaluation metrics for the Support Vector Machine (SVM) and Logistic Regression models, including accuracy, precision, recall, and F1-score. features a table that shows the precise metric values as well as a bar chart that visually compares the models' performance.



	Support Vector Machine	Logistic Regression
Accuracy	90.32%	91.74%
Precision	91.90%	90.86%
Recall	89.81%	94.01%
F1-Score	90.84%	92.41%

Figure 4-28

4.8.4 Saving the best model

joblib efficiently stored trained machine learning model by serializing it into a compact binary file, preserving its learned parameters and structure.

```python
import joblib
from sklearn.pipeline import Pipeline
# Save the trained model
joblib.dump(model, 'C:/Users/Ouma Becks/Desktop/Projects/Personal/Webpage/model.pkl')
```

```
['C:/Users/Ouma Becks/Desktop/Projects/Personal/Webpage/model.pkl']
```

Figure 4-29

4.8.5 Deploying the model

Deployed the model using Flask, creating a lightweight web application that exposes an API endpoint to receive input data, process it with the trained model, and return predictions in real-time.

```
PS C:\Users\Ouma Becks\Desktop\Projects\Personal\Webpage> & C:/ProgramData/anaconda3
/python.exe "c:/Users/Ouma Becks/Desktop/Projects/Personal/Webpage/app.py"
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use
 a production WSGI server instead.
 * Running on http://127.0.0.1:5000
```

Figure 4-30

4.8.6 Deployment interface

This interface detects network traffic anomalies by allowing manual input or file uploads, then processes the data through a trained model to predict and flag unusual patterns.

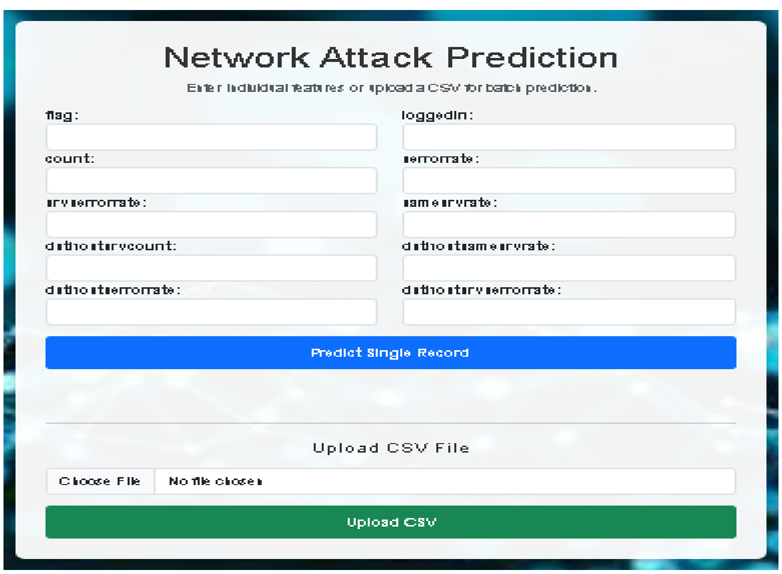


Figure 4-31

4.9 Conclusion

This chapter concluded with a thorough examination of machine learning approaches for network anomaly detection. To guarantee the best model performance, the dataset—which included 42 features and labeled traffic types—was pre-processed using encoding, feature selection, and normalization. Accuracy, precision, recall, F1-score, confusion matrices, and ROC curves were used to train and assess two supervised learning models: Logistic Regression and Support Vector Machine (SVM). The SVM model surpassed Logistic Regression with a higher accuracy of 95.14% and better F1-scores, demonstrating a stronger capacity to discern between malicious and normal traffic, even though both models showed significant classification abilities. These results were further corroborated by visualizations that showed improved anomaly score distributions and higher SVM AUC values. These findings highlight how well machine learning models—in particular, SVM—work to improve network security by accurately detecting anomalies.

CHAPTER FIVE
CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The goal of this study was to find out how well machine learning models operate at identifying network traffic irregularities, which is an essential task for preserving the security and integrity of computer networks. Through the use of a labeled dataset that included both harmful and legitimate network connections, this study effectively illustrated how supervised learning algorithms can accurately categorize and identify intrusions.

Managing categorical features, maintaining data consistency, and using feature selection strategies to find the most pertinent qualities for classification were all part of the data preprocessing phase. Training, testing, and evaluation were done on two machine learning models: Support Vector Machine (SVM) and Logistic Regression. The SVM model outperformed Logistic Regression, which had an accuracy of 91.56%, with a 95.14% accuracy rate. The models' performance was further evaluated using precision, recall, and F1-scores; SVM consistently produced superior results in recognizing both attack and regular traffic.

KDE plots, ROC curves, and confusion matrices were among the visualization tools that offered more in-depth understanding of the models' prediction power. Overall, the study demonstrated that SVM is a reliable and successful model for detecting network anomalies, with a high degree of confidence in differentiating between suspicious and regular activity.

5.2 Recommendations

Several suggestions are made for the future development and practical implementation of machine learning-based network anomaly detection systems in light of the study's findings:

Integration into Real-Time Monitoring Systems

To keep an eye on live traffic, the created models can be included into real-time intrusion detection systems (IDS). To do this, the models must be modified for streaming data and low-latency forecasts must be made.

Feature Engineering and Deep Learning

To automatically identify temporal and spatial patterns from raw network traffic data, advanced feature engineering or deep learning techniques such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) can be studied.

Continuous Learning and Adaptation

Deploying models that can continuously learn and adapt—such as through online learning or incremental learning techniques—will be crucial for long-term effectiveness since network traffic patterns change over time.

Security and Privacy Considerations

Preserving user privacy is essential while improving anomaly detection. Future systems ought to take into account federated learning and other privacy-preserving machine learning strategies.

5.3 Final Thoughts

One promising strategy for countering the growing threat of network assaults is machine learning. ML-based anomaly detection systems can be effective instruments in the continual endeavor to safeguard digital infrastructure if they are implemented, maintained, and improved upon. This study has established a strong basis for further investigation and implementation of automated and intelligent intrusion detection systems.

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APPENDICES

Budget

	Items	Description	Cost
1.	Hardware tools	Lenovo Thinkpad 8GB RAM Windows 11	Kshs.44,999
2.	Software tools	Python, Jupyter Notebook, SQL, Microsoft word, Flask and Github	Kshs.0
3.	Internet	Stable internet connectivity	Kshs.500
4.	Miscellaneous	Unforeseen expenses, Travell and communication costs and additional resources.	Kshs.3,000
5.	Write-up	Printing of the project proposal.	Ksh.250
		Total	Kshs.48, 749

Table 0-1

Work plan

Work plan

Weeks / Tasks	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Data collection												
Data preprocessing												
EDA												
Model training and selection												
Model testing and tuning												
Model evaluation												
Model deployment												

Table 0-2